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MOVIE TICKET BOOKING SYSTEM

VITyarthi Build Your Own Project

Python Console Application

1. Introduction

The Movie Ticket Booking System is a Python application designed to provide a structured workflow for movie selection, ticket booking, seat allocation, discounts and customer reward credits. The implementation uses modular Python files and JSON storage.

2. Problem Statement

Manual ticket booking may cause duplicate seat allocation, calculation mistakes and difficulty maintaining reward information. The project addresses these issues through validation, persistent storage and automated calculations.

3. Objectives

The objectives are to implement a real-world Python application, demonstrate modular programming, provide clear input/output, validate user input, store booking data, calculate offers correctly and demonstrate testing.

4. Functional Requirements

The system provides customer identification, movie selection, ticket quantity input, free-ticket processing, seat selection, Friday and bulk discounts, movie-credit management, booking storage and receipt generation. The project provides separate movie, credit, booking, storage and validation components.

5. Non-Functional Requirements

Usability: prompts and validation guide the user. Reliability: invalid input is handled without immediately terminating the application. Maintainability: responsibilities are separated into modules. Resource efficiency: JSON files are used for lightweight local persistence. Error handling is included throughout the input workflow.

6. System Architecture

The main controller coordinates validation, movie data, credit management and booking management. Storage provides a persistence layer for customer credits and booking records.

7. Design Diagrams

Architecture, workflow, use-case, sequence, class/component and storage/ER diagrams are provided in docs/diagrams.md using Mermaid notation.

8. Design Decisions and Rationale

Python was selected because the project applies programming fundamentals, functions, classes, data structures, file handling and exception handling. JSON was selected because the project is a small local application and does not require a database server. Dataclasses provide a clear representation for Movie and Booking objects.

9. Implementation Details

The source is divided into models.py, storage.py, validation.py, movies.py, credits.py, booking.py and main.py. This provides a modular implementation with meaningful files. The tests directory contains validation tests.

10. Screenshots / Results

For the final submission, run the program and capture screenshots of the movie list, seat selection, successful booking receipt and movie-credit output. These can be inserted into the report after execution.

11. Testing Approach

Tests cover generation of 50 seats, credit calculation for paid tickets and bulk-discount calculation. Manual validation also covers invalid movie numbers, invalid ticket quantities, duplicate seats, invalid seat names and empty customer names.

12. Challenges Faced

Important implementation challenges include keeping booking data persistent, preventing duplicate seat allocation, separating free and paid seats, handling reward credits and applying multiple discount rules without mixing the calculations.

13. Learnings and Key Takeaways

The project demonstrates modular design, classes and dataclasses, JSON file handling, input validation, exception handling, list/set operations, conditional logic, functions, testing and basic software documentation.

14. Future Enhancements

Possible future improvements include a graphical user interface, movie-show timing selection, payment integration, administrator controls, cancellation/refund support, database storage, password-based customer accounts and richer booking analytics.

15. References

Primary project requirements: VITyarthi Build Your Own Project - General Project Instructions & Submission Guidelines supplied for this project.

Screenshots & Results

Actual sample runs of the implemented Movie Ticket Booking System. The outputs below were captured from the Python application.

Sample Run 1 — 3 Tickets + Free Ticket

MOVIE TICKET BOOKING SYSTEM

Sample Run 1 — 3 Tickets + Free Ticket

```
==========================================================
MOVIE TICKET BOOKING SYSTEM
==========================================================

Enter your name:
Welcome, Ved Demo!

Available Movies
----------------------------------------------------------
1. Avengers: Endgame - Rs.250.00
2. Interstellar - Rs.300.00
3. Inception - Rs.280.00
4. Spider-Man: No Way Home - Rs.220.00

Enter movie number:
Enter number of tickets:
You need to select 4 seats.
Available Seats:
A6, A7, A8, A9, A10, B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, D1, D2,

Enter 4 seat numbers separated by spaces:
==========================================================
BOOKING CONFIRMED
==========================================================
Booking ID    : MOV88495
Name          : Ved Demo
Movie         : Avengers: Endgame
Ticket Price  : Rs.250.00

Paid Tickets
Number        : 3
Seats         : A6, A7, A8

Free Tickets
Seats         : A9

----------------------------------------------------------
Subtotal       : Rs.750.00
Total Discount : Rs.0.00
Final Amount   : Rs.750.00

==========================================================
MOVIE CREDIT SYSTEM
==========================================================
Credits Earned : +45
Total Credits  : 45
==========================================================
Thank you for booking with us!
==========================================================
You need 45 more credits for a free ticket.
```

Sample Run 2 — 5 Tickets + Bulk Discount

MOVIE TICKET BOOKING SYSTEM

Sample Run 2 — 5 Tickets + Bulk Discount

```
==========================================================
MOVIE TICKET BOOKING SYSTEM
==========================================================

Enter your name:
Welcome, Bulk Demo!

Available Movies
----------------------------------------------------------
1. Avengers: Endgame - Rs.250.00
2. Interstellar - Rs.300.00
3. Inception - Rs.280.00
4. Spider-Man: No Way Home - Rs.220.00

Enter movie number:
Enter number of tickets:
You need to select 6 seats.
Available Seats:
A6, A7, A8, A9, A10, B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, D1, D2,

Enter 6 seat numbers separated by spaces:
==========================================================
BOOKING CONFIRMED
==========================================================
Booking ID    : MOV27238
Name          : Bulk Demo
Movie         : Interstellar
Ticket Price  : Rs.300.00

Paid Tickets
Number        : 5
Seats         : B1, B2, B3, B4, B5

Free Tickets
Seats         : B6

----------------------------------------------------------
Subtotal       : Rs.1500.00
Bulk Discount  : -Rs.225.00
Total Discount : Rs.225.00
Final Amount   : Rs.1275.00

==========================================================
MOVIE CREDIT SYSTEM
==========================================================
Credits Earned : +75
Total Credits  : 75
==========================================================
Thank you for booking with us!
==========================================================
You need 15 more credits for a free ticket.
```

Sample Run 3 — 90-Credit Free Ticket

MOVIE TICKET BOOKING SYSTEM

Sample Run 3 — 90-Credit Free Ticket

```
==========================================================
MOVIE TICKET BOOKING SYSTEM
==========================================================

Enter your name:
Welcome back, Credit Demo!
You have 90 movie credits.

You have enough credits for a free ticket. Use 90 credits? (yes/no): Free ticket added!

Available Movies
----------------------------------------------------------
1. Avengers: Endgame - Rs.250.00
2. Interstellar - Rs.300.00
3. Inception - Rs.280.00
4. Spider-Man: No Way Home - Rs.220.00

Enter movie number:
Enter number of tickets:
You need to select 2 seats.
Available Seats:
A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, C1, C2, C3, C4, C5, C6, C7, C

Enter 2 seat numbers separated by spaces:
==========================================================
BOOKING CONFIRMED
==========================================================
Booking ID    : MOV61718
Name          : Credit Demo
Movie         : Inception
Ticket Price  : Rs.280.00

Paid Tickets
Number        : 1
Seats         : C1

Free Tickets
Seats         : C2

----------------------------------------------------------
Subtotal       : Rs.280.00
Total Discount : Rs.0.00
Final Amount   : Rs.280.00

==========================================================
MOVIE CREDIT SYSTEM
==========================================================
Credits Earned : +15
Total Credits  : 15
==========================================================
Thank you for booking with us!
==========================================================
You need 75 more credits for a free ticket.
```